

SATD Detection: Qwen3 vs E5 vs BGE-M3 Embeddings

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1 What we did

We compared three modern text-embedding models — **Qwen3-Embedding-0.6B**, **E5-Large-v2**, and **BAAI/bge-m3** — for classifying Self-Admitted Technical Debt (SATD). This report extends the earlier Qwen3-vs-E5 study with a full experiment using BGE-M3’s dense embeddings, run through the identical pipeline. Texts were embedded by mean-pooling and L2-normalizing the encoder outputs, and a two-step classification scheme was applied per artifact type (code comments, issues, commit messages, pull requests):

1. **Identification** — binary: SATD vs. Not-SATD.
2. **Categorization** — four-class: C/D, REQ, TES, DOC (SATD rows only).

Each task was trained with **XGBoost** and **Logistic Regression** on each embedding set, giving six combinations per artifact per task. Results were compared against the paper’s baselines (BiLSTM+AugGPT for identification; BERT+AugGPT for categorization), with the supervisor’s replication numbers used for commit messages.

All three experiments used the paper’s pre-augmented dataset ($\approx 109k$ rows; 95,704 after cleaning), which equalizes the four SATD subtypes per artifact while retaining a Not-SATD majority (65–82%). The train/validation/test split (80/10/10, stratified) was performed *before* augmentation to avoid leakage, and Logistic Regression was run with $C = 0.5$ throughout. Only the embedding model differed between runs.

2 Results

Tables ?? and ?? report macro-F1 on the test sets for each embedding with XGBoost (the classifier that consistently outperformed logistic regression), alongside the baselines.

Table 1: Identification (SATD vs. Not-SATD), macro F1, XGBoost.

Artifact	Baseline	Qwen3	E5	BGE-M3
Code comments	0.939	0.9664	0.9587	0.9594
Issues	0.878	0.8656	0.8632	0.8533
Pull requests	0.862	0.8642	0.8703	0.8580
Commit messages	0.910	0.8917	0.8817	0.9014

Key findings:

Table 2: Categorization (C/D, REQ, TES, DOC), macro F1, XGBoost.

Artifact	Baseline	Qwen3	E5	BGE-M3
Code comments	0.882	0.9470	0.9581	0.9558
Issues	0.899	0.9442	0.9503	0.9562
Pull requests	0.876	0.9548	0.9335	0.9644
Commit messages	0.980	0.9639	0.9487	0.9603

- **BGE-M3 is the best categorizer.** It produced the highest macro F1 on three of four artifacts (issues, pull requests, and joint-best on code comments) and the largest margin over the BERT baseline anywhere in this study: +0.0884 on pull requests. Its categorization average with XGBoost (0.9592) beats Qwen3 (0.9525) and E5 (0.9477).
- **BGE-M3 handles commit messages best.** It was the only embedding within one point of the BiLSTM baseline on commit identification (0.9014 vs. 0.9100), ahead of Qwen3 (0.8917) and E5 (0.8817).
- **Identification is close among all three.** With XGBoost the averages are Qwen3 0.8970, E5 0.8935, BGE-M3 0.8930 — a spread of under half a point, with Qwen3 winning on code comments and issues and E5 on pull requests.
- **BGE-M3 is the worst fit for logistic regression.** With LogReg it fell to 0.8407 (identification) and 0.8571 (categorization) on average — below E5 and Qwen3 and, in categorization, roughly 0.03–0.05 below its own XGBoost results.
- **XGBoost consistently beat Logistic Regression for every embedding** (e.g., BGE-M3: 0.9261 vs. 0.8489 combined average), confirming the earlier finding.
- **Best overall combination: BGE-M3 + XGBoost** (0.9261 averaged over both tasks), narrowly ahead of Qwen3 + XGBoost (0.9248) and E5 + XGBoost (0.9206).
- **Commit messages remain the hardest artifact:** no embedding reached the categorization baseline (0.980) on commit messages, and all were below it on commit identification.

3 Takeaways

1. **BGE-M3 is the strongest choice for SATD categorization**, beating fine-tuned BERT on three of four artifacts and edging out Qwen3 and E5, especially on issues and pull requests. Its dense embeddings encode debt-type distinctions particularly well.
2. **For identification the three models are interchangeable** (all within 0.4 points with XGBoost); BGE-M3’s best showing on commit messages makes it a slight favorite where that artifact matters.
3. **Classifiers dominate embedding choice.** The gap between the best and worst embedding is far smaller than the gap between XGBoost and logistic regression — embedding selection matters less than classifier selection. BGE-M3 in particular rewards tree-based classifiers and loses its advantage with a linear head.

4. **The paper’s identification results remain tied to its own augmented data**, not reproduced by simply swapping the embedding model: no embedding matched the BiLSTM baseline on issues or commit messages, and only Qwen3 and E5 did on code comments and pull requests, respectively.
5. **Recommendation for this task: BGE-M3 embeddings + XGBoost**. It gives the best categorization performance and the best commit-message identification, with no tuning cost beyond a standard classifier.